The first later cut when the adjacent suffix is intransitive

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses and acquired proofs: $G \le A_n$ is transitive with a connected string C-marking, all maximal omissions are intransitive, and the actual first three marks give an extendable six-point head on D. In the contrary range $n = 2r \ge 18$, write

$$P_k = G_{\le k}, \qquad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus (D \setminus \{6\})).$$

The actual head orbit C_k of P_k carries its pure alternating kernel, with the complete acquired parity-fibre formula and strict growth. In this note impose the extra hypothesis

$$B = G_{\ge 4} \text{ is intransitive on } \Omega \setminus D. \tag{1}$$

Eligibility always refers to a full ambient omission.

Theorem 1. *Let $5 \le j \le r-2$ be the least later label with a unique eligible edge. Suppose this cut has a first shared label $g < j$ with $g \ge 5$, and $P = P_{g-1}$ is intransitive on its support Y. Then all of the following hold.*

1. *P has precisely two nontrivial orbits: its head orbit C_{g-1} and one exterior orbit. Moreover $|Y| = 2g+1$.*

2. *$j = g+1$. Write the two G_j-orbits as $C = C_{j-1}$ and $A = \Omega \setminus C$, with crossing endpoint $b \in C$. Then*

$$|A| = 2(r-g-1), \qquad \mathrm{Fix}(P) = A \cup \{b\}, \qquad C = C_{g-1} \,\dot\cup\, E \,\dot\cup\, \{b\},$$

   where E is the exterior P-orbit.

3. *Put $Q = G_{>j}$ and $t = r-g-1$. If Q is transitive on A, it has degree $2t$ and rank $t-1$ there. If it is intransitive on A, it has two faithful orbits of size t, and its action on each is the natural S_t.*

4. *The actual interval $H = G_{\{3,\dots,g-1\}}$ has two nontrivial orbits, $C_{g-1} \setminus (D \setminus \{6\})$ and E. Its paired ambient interval bound is an equality:*

$$2(g-3) = |\operatorname{supp}(H)| - 2.$$

   A minimum-square selection has exactly one tree component in each of these two orbits; square components are not excluded.

Proof. The complete early-cut calculation gives at least two eligible edges at label 3 under (1), and at label 4 in every critical head. Minimality of j gives at least two for every label $5, \dots, j-1$: at least one exists because the full group is transitive and its omission is intransitive. Hence every label $3, \dots, g-1$ is paired in the ambient sense.

The acquired unique-cut orbit lemma identifies the head-side G_j-orbit as C_{j-1} and shows that Q fixes it pointwise. Its other orbit A is nontrivial, because a singleton admits no shared label. The hypothesis $j \le r-2$ makes the reversed crossing interior, as required by these support theorems. The full reversed transitive-side or intransitive-side support theorem therefore applies. It gives

$$|Y| \le 2g+1. \tag{2}$$

This is its actual-support conclusion, not a bound on an arbitrary projected group.

Let p be the number of nontrivial P-orbits, so $p \geq 2$. The exact prefix intersection formula, using actual ambient IP, identifies $H = P \cap R$. It removes exactly the five head points $D \setminus \{6\}$ from the support and from the head orbit, and leaves every exterior orbit unchanged. Since $g \geq 5$, the remaining head orbit has at least three points and is nontrivial. Thus

$$|\operatorname{supp}(H)| = |Y| - 5, \qquad o^*(H) = p,$$

where o^* counts nontrivial orbits. The full acquired ambient interval theorem gives

$$2g - 6 = 2|\{3, \ldots, g-1\}| \leq |Y| - 5 - p \leq 2g - 4 - p. \tag{3}$$

It follows that $p \leq 2$. Hence $p = 2$, and both inequalities in (3) are equalities. This proves the first and fourth conclusions, including the tree-component count from equality in that theorem.

Now (2) is an equality. Apply the complete equality interfaces of the reversed support theorems, with reversed crossing $i' = r - 1 - j$ and last shared index $h' = r - 1 - g$. In the transitive Q case, they give $h' = i' + 1$, $|A| = 2h'$, and exact fixed set $A \cup \{b\}$. In the intransitive Q case, strictness unless $h' = i' + 1$ gives the same index equality; equality forces two faithful Q-orbits of common size $i' + 1$, natural symmetric action on each, and the same exact fixed set. Thus in both cases $j = g + 1$, $h' = t$, and $|A| = 2t$. The literal suffix rank is $r - 1 - j = t - 1$. The exact fixed set and the two nontrivial P-orbits give the asserted partition of C. No conclusion that Q is natural alternating, or that its transitive case is already classified, has been inserted. □

Proposition 2 (The bound without adjacent-suffix transitivity information)**.** *Retain the theorem's interior-cut, minimality, $g \geq 5$ and intransitive-prefix hypotheses, but drop (1). Then P_{g-1} has at most three nontrivial orbits. If it has three, then $|Y| = 2g+1$, $j = g+1$, and the same reversed-side equality conclusions hold. Moreover adjoining ρ_3 to $K = G_{\{4,\ldots,g-1\}}$ merges exactly two of its point orbits and leaves every other orbit unchanged.*

Proof. The label 4 and all labels $5, \ldots, g-1$ remain paired; no assertion about label 3 is needed. Write $d(V) = n - o(V)$ for the point-orbit deficit of a subgroup V. The acquired ambient interval theorem gives $d(K) \geq 2(g-4)$. Adjoining ρ_3 strictly changes the orbit partition. Otherwise K and $H = \langle K, \rho_3 \rangle$ would have the same point orbits; adjoining all other ambient marks would then give identical orbit partitions for G_3 and G, contradicting the intransitive omission. Hence

$$d(H) \geq d(K) + 1 \geq 2g - 7.$$

The exact prefix intersection still gives $d(H) = |Y| - 5 - p$, where p counts the nontrivial P-orbits. With (2) this yields $p \leq 3$. At $p = 3$ all inequalities are equalities. In particular $|Y| = 2g + 1$ and the same reversed interior equality interfaces apply. The equality $d(H) - d(K) = 1$ says precisely that the number of point orbits decreases by one; as adjoining a generator can only merge orbits, exactly two old orbits merge. □

Attribution and scope. The new step is to combine the complete acquired actual prefix intersection with the acquired ambient interval deficit before using the reversed equality interfaces. Their full proofs remain paid, including all symmetric-string inputs. The underlying source route is L044 Propositions 5.20–5.32. This is a conditional reduction for all g at once; it does not eliminate the resulting configuration. Endpoint cuts $j = r - 1$ remain separate: the reversed interior equality interfaces used here do not apply to them. The adjacent suffix hypothesis (1) is essential for the paired label 3. The case where P is transitive on its support is governed separately by the reviewed exceptional-end alternative. The final proposition gives a weaker three-orbit bound without (1). No whole-original, question, formal or numerical credit is claimed.